



Effects of Artificial Night Lighting on Fireflies: Global Synthesis of Scientific Evidence

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Abstract: Artificial light at night (ALAN) has become a problem for fireflies because it disrupts their natural processes and threatens the conservation of their populations. In this regard, the aim of the study was to determine the effects of ALAN on firefly species through a systematic review. The PRISMA 2020 statement was fundamental for the review of the databases, and the inclusion and exclusion criteria for specifying the subject of study. On the other hand, the annual growth of scientific production was determined using the digital tool (Calcuvio). The year and country with the highest scientific production were 2021 and the United States, respectively, and the annual growth (2005–2025) was 16%. The most studied species was *Lampyrus noctiluca*, and the effect of ALAN on the most common fireflies was a change in the intensity and frequency of their flashes in females. It is concluded that investment should be made in research in countries with abundant and diverse populations of fireflies. Furthermore, studies should be conducted on trophic interactions or sublethal physiological effects of fireflies, as well as on diversifying the species under study.

Keywords: Bioluminescence; Light pollution; Conservation; Ecosystems; Insects

1 Introduction

Natural landscapes are increasingly affected by human activities, light pollution, habitat destruction, and noise, which have become more frequent and intense [1].

Artificial light at night (ALAN), such as that from vehicles, streets, security, and businesses [1, 2], has steadily increased by 10% annually in intensity in the night sky over the last decade [3–6]. This has resulted in greater light intensity (10–60 lx) compared to natural nighttime light sources, such as the moon (0.1–0.6 lx) [7].

ALAN is a global threat because it has far-reaching repercussions on many species of nocturnal wildlife [3, 8, 9], with the most scientific documentation on bats, insects, and birds [7].

Fireflies (Coleoptera: Lampyridae) are found throughout the world, but most of their diversity is in the Neotropics and Southeast Asia [10]. They have also become a nighttime ecotourism activity and are considered bioindicators of ecosystem health [11]. ALAN, along with pesticide use and habitat loss, has been found to be a detrimental factor contributing to the decline of the global firefly population [12–15], as fireflies rely on bioluminescent signals to communicate, making them vulnerable to interference from ambient light [16]. Bioluminescent signals provide information about mate quality, sex, and species identity [13]. When exposed to ALAN, female fireflies glow less brightly and attract fewer males [17–20], affecting courtship (locating mates) and predation by inhibiting glow or flash behavior [21, 22].

This review study identified the types of ALAN and their effects on fireflies (Coleoptera: Lampyridae) in different species. In this regard, the following research questions were formulated: In which year and in which country is scientific production highest? What is the annual growth in scientific production? Which species is the most studied? And what effect of ALAN is most frequently observed in the studies reviewed?

2 Methodology

The PRISMA 2020 statement [23] was used to identify, select, and analyze articles from scientific journals [24, 25]. The initial search was conducted in June 2025 and involved identifying relevant literature in Scopus, Taylor & Francis, Wiley, and Ebsco using the search equation: ((“light pollution” OR “artificial light at night” OR ALAN OR “night-time lighting” OR skyglow) AND (firefly OR fireflies OR “Lampyridae” OR “glow-worm” OR “luciernaga” OR “luciérnaga”) AND (impact* OR effect* OR behavi* OR courtship OR “mating success” OR bioluminesc* OR ecology OR population* OR dispersal)). And on ScienceDirect with: (“light pollution” OR “artificial light at night” OR ALAN) AND (firefly OR fireflies OR “Lampyridae” OR “glow-worm”). The databases used for the search are recognized in the international scientific community, covering publications from around the world for the collection of information.

Next, articles related to the topic in the title, abstract, and keywords were selected for further analysis. The inclusion criteria covered: (1) scientific research articles, (2) articles worldwide, (3) articles in all languages, and (4) articles published in all years up to June 2025. On the other hand, the following were excluded: (1) duplicate articles between databases, (2) articles that require payment to access the information, (3) articles not related to the research objective, and (4) other scientific sources (conference papers, reviews, books, among others).

A group of two authors were responsible for searching the database, and any conflicts were resolved at the end of the search with the lead author (M.R.-I.). To organize the articles, we used the online tool for creating flowcharts, PRISMA 2020 [26]. Initially, 2,015 articles were identified, and after applying the inclusion and exclusion criteria, 23 remained for review (Figure 1).

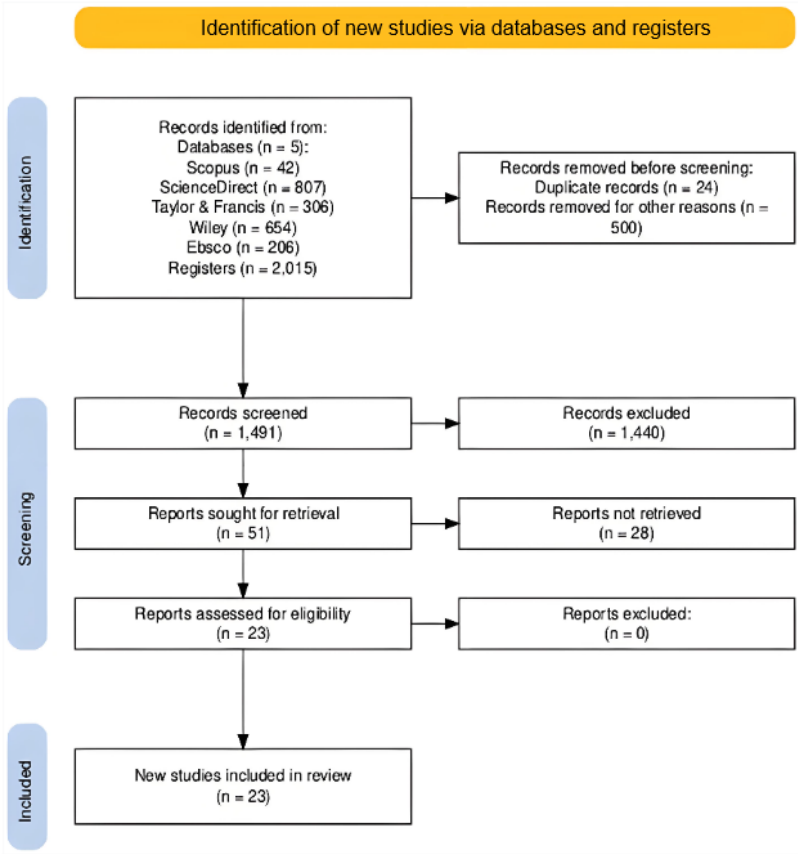


Figure 1. Study selection diagram

2.1 Compound Annual Growth Rate

The annual growth in scientific output between 2004 and 2025 was determined using the following equation [27]:

$$CAGR\% = 100 * \left(\left(\frac{V_f}{V_i} \right)^{\frac{1}{t}} - 1 \right)$$

where,

$CAGR$ = Compound Annual Growth Rate

V_i = Initial value

V_f = Final value

t = Years

$$CAGR\% = 100 * \left(\left(\frac{23}{1} \right)^{\frac{1}{21}} - 1 \right)$$

$$CAGR\% = 16$$

2.2 Data Analysis

The information was downloaded from the databases in CSV format to analyze and determine the distribution of studies by year and country. Subsequently, the information was disaggregated by light type, species, variable measured, and effects.

3 Results and Discussions

The high number of studies conducted in 2021 (Figure 2) on the effects of artificial night light on fireflies responds to the convergence of scientific milestones and conservation policies that highlighted the problem and raised new and urgent questions. In 2021, the first assessments of the risk of extinction of fireflies in North America were published, identifying light pollution, habitat loss, and climate change as the main threats [28]. In this regard, given that the first evidence emerged in early 2021, it is possible that research on the subject has begun, reflecting increased scientific output. Likewise, in that year, sustainable firefly tourism generated management guidelines and standards that indicated the reduction of ALAN as a conservation requirement [29, 30], thereby increasing the demand for evidence on lighting thresholds and wavelength spectra, which led to further research. On the other hand, the anthropause generated by COVID-19 (2022 and 2023) has led to “natural experiments” being carried out without the intervention of ALAN [31–33], which has led to an increase in studies.

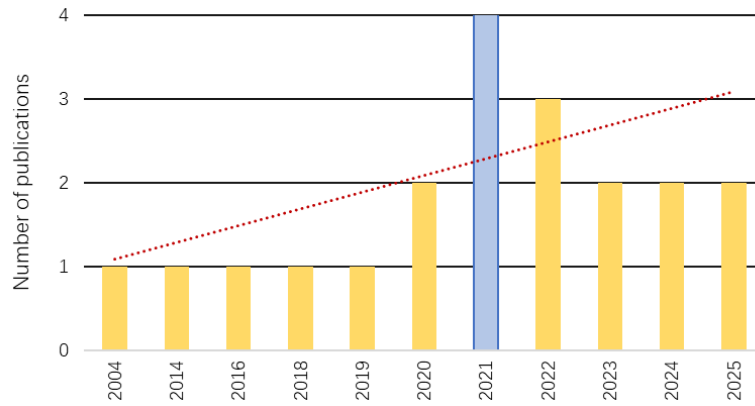


Figure 2. Scientific output per year

The scarcity of scientific publications prior to 2004 is consistent with the historical trajectory in this field, as studies on the effects of artificial nighttime light on fireflies have been conducted over the last two decades [16]. These have been driven by the rapid expansion of more efficient and broader spectrum lighting technologies, particularly light-emitting diodes (LED) [34, 35]. On the other hand, the compound annual growth rate of scientific output was 16%, indicating that it continues to be an emerging field in recent years.

The analysis shows that the United States accounts for the largest share of scientific output with seven studies (Figure 3), reflecting a recurring trend in environmental sciences. Furthermore, the United States leads the way in knowledge generation in multiple areas due to its strong funding for research, academic infrastructure, and stable global collaboration networks [36–38]. This leadership, however, contrasts with the geographical distribution of fireflies, where Latin America has lower scientific output [10, 39], as is the case in Brazil and Colombia, with two and one study, respectively. This imbalance indicates that most of the published scientific evidence mainly reflects the

interaction of the effects of artificial night light on fireflies in the northern hemisphere. This does not reflect the actual distribution of firefly biodiversity, as the highest levels of richness and endemism are concentrated in tropical regions of the southern hemisphere, specifically in Latin America and Southeast Asia. This bias is due to structural factors in scientific research, such as greater funding, stable academic infrastructure, and long-term monitoring programs in countries in the northern hemisphere, such as Europe and North America. Furthermore, firefly species in the northern hemisphere, such as *Lampyris noctiluca*, have well-documented life cycles and easily observable signaling behaviors, making them accessible experimental models for studies on light pollution. In contrast, in megadiverse regions of the global south, research is limited by logistical difficulties, reduced access to spectral measurement technologies, and a scarcity of detailed taxonomic inventories.

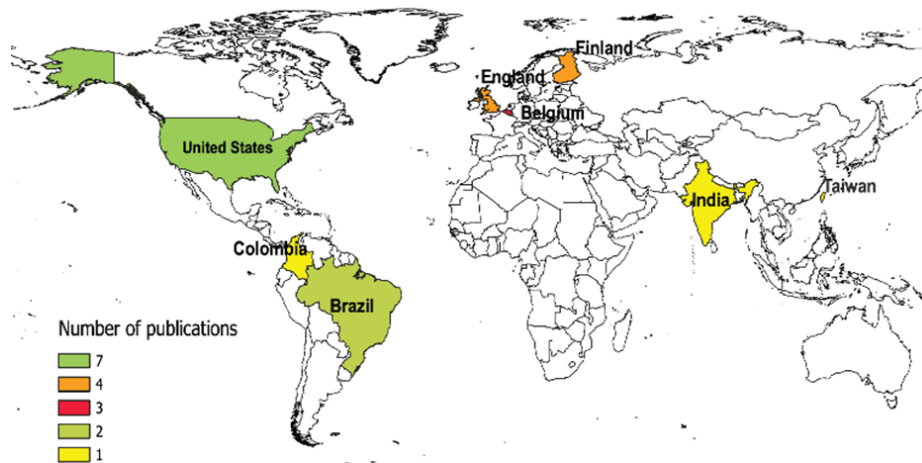


Figure 3. Scientific output by country

The preponderance of *Lampyris noctiluca*, or the common European firefly, in studies on the effects of ALAN is due to the fact that it is a widely distributed and prominent species in Europe whose reproduction depends heavily on the sedentary bioluminescence of females to attract flying males, which creates a natural system that can be observed and manipulated experimentally when investigating light interference in sexual communication [16, 17].

Experimental field research reveals that in *Lampyris noctiluca*, the presence of ALAN decreases mating success and alters mating search behaviors [18, 20, 40, 41]. This would explain why researchers interested in studying the behavioral and demographic effects of ALAN have frequently used *Lampyris noctiluca* as a model species. Likewise, it is a concern for conservationists around the world. Scientific evidence and announcements from conservation organizations have highlighted the decline and threatened status of lampyridae species, and in some reports, *Lampyris noctiluca* appears as a taxon at increasing risk in landscapes with ALAN [42, 43]. This need for evidence for management and conservation motivates research aimed at understanding the relationship between ALAN and population dynamics in this species specifically. From a practical perspective, the biology of *Lampyris noctiluca* favors its use in experimental research: its seasonal activity is predictable, females emit static signals that are easy to locate and quantify, and males respond with a search pattern that can be observed and manipulated in field and laboratory conditions [14]. These characteristics reduce methodological complexity compared to other species of fireflies that exhibit more mobile behaviors or more complex signaling systems.

LED light is the most widely used source in the studies reviewed, due to its use in public lighting [44, 45], and its spectral and control properties facilitate experimentation [46, 47]. Likewise, its rapid increase in the extent and intensity of light pollution worldwide [48, 49], makes it the main source of negative effects on nocturnal insects, especially bioluminescent species such as fireflies [50–52]. On the other hand, since 2011, sky brightness has been increasing by between 7% and 11% annually, placing LED lighting at the center of the problem [49, 53].

The predominant effects in the ALAN studies reviewed on fireflies was the modification of the intensity and frequency of their flashes in females (Table 1), which prevents them from participating in courtship (they do not attract males) and, consequently, affects reproduction, as bioluminescent signals are extremely sensitive to artificial light.

Experimental studies have shown significant reductions in flash activity in both males and females exposed to different types and intensities of artificial light, which directly decreases the likelihood of successful courtship interactions [54, 55].

ALAN affects the light signals of fireflies through several mechanisms: first, artificial light in the environment prevents males from distinguishing the flashes of females, as the bright background interferes with their natural signal [16]. Second, exposure to artificial light alters flashing behavior (reducing the flashing rate or changing the

synchrony between individuals) [20]. On the other hand, it is not only light intensity that is important; the spectrum and duration of illumination also determine the degree of disturbance. Research has shown that wavelengths close to the peak of bioluminescence (intense amber lights) can interrupt, alter, or block sexual communication (courtship and mating) [17, 56]. Overall, the studies reviewed show a consistent pattern of spectral sensitivity in fireflies to ALAN. Short wavelengths (blue and white) generate the most disruptive effects, significantly reducing flashing activity, mating success, and reproduction. In contrast, long wavelengths (both blue and red) tend to produce less severe effects, although they are not completely harmless, and their impact depends on the intensity and duration of exposure. This highlights that the spectral overlap between artificial light and the visual sensitivity range of fireflies is an important factor in the interference of bioluminescent communication.

Table 1. Effects of Artificial light at night (ALAN) on fireflies

Ref.	Type of Light	Species	Measured Variable	Effects
[57]	Green LED light (550 nm)	<i>Lampyris noctiluca</i>	Vision	Distinguishes light signals based on chromaticity.
[58]	ALAN	<i>Lampyris noctiluca</i>	Phototaxis	Interference with phototaxis; males cannot detect females for mating.
[59]	LED light	^a <i>Photuris versicolor</i> ; ^b <i>Photinus pyralis</i>	Abundance, dispersion, mating activity	^a Reduced flash capacity; ^b Reduction in courtship behavior and reproductive success.
[60]	LED light	<i>Aquatica ficta</i>	Intensity and frequency of flashes	Modification of flash intensity and frequency within species' visual range.
[22]	LED light	^a <i>Photuris versicolor</i> ; ^b <i>Photinus pyralis</i>	Attraction/repulsion and courtship	^a Attracted to ALAN, less courtship; ^b Reduced reproductive success.
[18]	LED light	<i>Lampyris noctiluca</i>	Attraction to males	Females decrease success of attracting a mate.
[19]	White LED light	Not specified	Attraction to males	Prevents females from attracting males.
[56]	Amber LED light	<i>Photinus obscurellus</i>	Intensity and frequency of flashes	Changes in male flash frequency and intensity.
[61]	ALAN	<i>Amydetes fastigiata</i>	Stress	Stress reduces growth rate and tracking of females.
[62]	White LED light	^a <i>Photuris</i> sp.; ^b <i>Photinus obscurellus</i>	Food	^a Weight gain in larvae; During nighttime feeding by burrowing into the soil rather than dispersing. ^b No effects.
[40]	White LED light	<i>Lampyris noctiluca</i>	Location	Affects males locating females.
[63]	White LED light	<i>Aquatica ficta</i>	Larval development	High larval mortality.
[20]	ALAN	<i>Lampyris noctiluca</i>	Reproduction	Reduced reproductive success.
[64]	White LED light	<i>Lampyris noctiluca</i>	Sexual signaling	Females stop shining.
[65]	Blue LED, broad-spectrum amber, red	<i>Photinus carolinus</i>	Flight and courtship behavior	Attraction to blue and amber; greater courtship under blue and red.
[55]	White LED light	^a <i>Photinus obscurellus</i> ; ^b <i>Photinus pyralis</i> ; ^c <i>Photinus marginellus</i> ; ^d <i>Photinus greeni</i>	Mating and movement	^a No mating; ^{b,c} Reduced reproductive success; ^d Difficulty locating partner.

Ref.	Type of Light	Species	Measured Variable	Effects
[16]	ALAN	Not specified	Diversity and distribution	Reduced diversity and distribution in high-ALAN areas.
[17]	Blue LED light (452 nm), white (449 nm), yellow (575 nm), or red (625 nm)	<i>Lampyrus noctiluca</i>	Attraction to males	Mate attraction success was wavelength-dependent, with short wavelengths (blue and white) being more detrimental than long ones (yellow and red).
[41]	Green LED light	<i>Lampyrus noctiluca</i>	Location	Reduced probability of males locating females.
[54]	White LED light	<i>Photinus</i> and <i>Photuris</i>	Sparks	Significant reduction in flashing during courtship and feeding.
[66]	ALAN	<i>Abscondita chinensis</i> ; <i>Abscondita perplexa</i> ; <i>Abscondita terminalis</i>	Abundance	Decrease in larvae and adults.
[67]	White LED light (peak 450 nm, range 410–740 nm), yellow (peak 587 nm, range 490–740 nm), and red (peak 617 nm, range 555–740 nm)	<i>Lampyrus noctiluca</i>	Phototactic behavior of males	Positive phototaxis to yellow and red; negative to white; the latter also lowering activity.
[68]	Red, white and blue LED	<i>Lampyrus noctiluca</i>	Larval movement	No phototaxis to ALAN was observed in either sex. However, larval activity was suppressed by short-wavelength (blue and white) light, but not by red.

4 Conclusions

It is suggested that greater investment in research should be encouraged in countries with greater abundance and diversity of fireflies (Colombia, Peru, Brazil, Mexico, Thailand, Malaysia) is emphasized, as well as strengthening transnational collaborations to balance scientific production and promote conservation strategies. Likewise, scientific production on the subject will increase progressively. On the other hand, the concentration of studies on *Lampyrus noctiluca* reveals biases and gaps in the literature. Many regions worldwide (Asia, tropical America, and Africa) are underrepresented, with most studies focusing on immediate behavioral effects (interference with signaling and mating) and less on long-term demographic consequences, trophic interactions, or sublethal physiological effects (stress, changes in circadian rhythms, larval mortality).

Recent reviews recommend diversifying the species studied, expanding temporal and spatial scales, and combining experimental studies with population monitoring to translate observed effects into population viability forecasts. From a conservation and management perspective, clear and enforceable measures are recommended, such as reducing lighting intensity in fragile habitats, using shielding and beam direction, and selecting less disruptive spectra outside the range of maximum sensitivity of fireflies. Likewise, in protected natural areas, lighting should be minimal or non-existent (≤ 1 lx), prioritizing ecological darkness. In urban coastal areas, it is necessary to reduce light intensity (5–10 lx), use warm lights, and reduce hours of operation in order to mitigate impacts without compromising urban safety. At ecotourism sites, very low and controlled lighting (≤ 3 lx) with restricted illumination and non-invasive spectra is recommended, allowing tourism to be compatible with conservation.

Finally, to strengthen conclusions in future research, it is suggested that controlled experiments be conducted to quantify the intensity and spectral bands that affect courtship behaviors. It is also recommended that field monitoring be combined with medium-term population data and that socio-environmental studies be integrated to evaluate the practical possibilities for mitigation in urban or ecotourism areas.

5 Limitations

The limitations encountered in the review were: the concentration of studies in certain regions (northern hemisphere), the focus on a limited set of species (*Lampyris noctiluca*), the predominance of short-term studies, and the lack of long-term population data.

Author Contributions

Conceptualization, M.R.-I. and R.P.D.; methodology, Y.Q.C.; software, C.A.N.T.; validation, A.A.V. and J.K.G.L.; formal analysis, J.K.G.L.; investigation, R.R.-I.; resources, Z.R.-V.; data curation, Y.Q.C.; writing—original draft preparation, M.R.-I., R.R.-I., and Y.Q.C.; writing—review and editing, M.R.-I., R.P.D., and Y.Q.C.; visualization, Z.R.-V.; supervision, M.R.-I.; project administration, M.R.-I. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

Conflicts of Interest

The authors declare that they have no conflicts of interest.

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